Math3707 Cosmology CW

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Question 1

Part a

$$H_0 = \frac{\dot{a}(t_0)}{a(t_0)} \tag{1}$$

Where H_0 is the Hubble constant.

$$q_0 = -\frac{\ddot{a}(t_0)}{a(t_0)H_0^2} = -\frac{\ddot{a}(t_0)a(t_0)}{\dot{a}(t_0)^2} \tag{2}$$

Where q_0 is the deceleration parameter.

Part b

Let $f(t) = \frac{1}{a(t)}$, then Taylor expanding $f(t)$:

$$\begin{aligned} f(t) &= f(t_0) + (t-t_0)f'(t_0) + \frac{(t-t_0)^2}{2}f"(t_0) \\ f(t) &= \frac{1}{a(t_0)} - \frac{(t-t_0)\dot{a}(t_0)}{a(t_0)^2} + \frac{(t-t_0)^2}{2}\left(\frac{2\dot{a}(t_0)^2}{a(t_0)^3} - \frac{a(t_0)\ddot{a}(t_0)}{a(t_0)^3}\right) \\ a(t_0) &= 1 \\ \therefore f(t) &= 1 - \tau H_0 + \frac{\tau^2}{2}\left(2H_0^2 + q_0H_0^2\right) \\ f(t) &= \frac{1}{a(t)} = 1 - \tau H_0 + \frac{\tau^2 H_0^2}{2}(2+q_0) \end{aligned} \tag{3}$$

Using the Taylor expanded form of $\frac{1}{a(t)}$:

$$\chi = c\int_t^{t_0} \frac{1}{a(t)}dt \tag{4}$$

Performing a change of variables in (4):

$$\begin{aligned} \tau = t - t_0 \therefore \frac{d\tau}{dt} = 1 \Rightarrow d\tau = dt \\ \text{lower limit:} t = t \Rightarrow \tau = t - t_0 = \tau \\ \text{Upper limit:} t = t_0 \Rightarrow \tau = t_0 - t_0 = 0 \end{aligned} \tag{5}$$

Applying the change of variables:

$$\begin{aligned} \chi &= c\int_\tau^0 1 - \tau H_0 + \frac{\tau^2 H_0^2}{2}(2+q_0)dt \\ \chi &= c\left[\tau - \frac{H_0}{2}\tau^2 - \frac{H_0^2}{6}(2+q_0)\tau^3\right]_\tau^0 \\ \chi &= c\left(-\tau + \frac{H_0}{2}\tau^2 - \frac{H_0^2}{6}(2+q_0)\tau^3\right) \\ &\text{neglecting } \tau^n, n > 2 \text{ terms:} \\ \chi &= c\tau\left(\frac{H_0\tau}{2} - 1\right) \end{aligned} \tag{6}$$

Part c

$$\begin{aligned} H_0\tau &= -z + (2+q_0)\frac{z^2}{2} \\ \Rightarrow \tau &= \frac{-z + 0.5(2+q_0)z^2}{H_0} \\ \tau &= \frac{Z}{H_0} \end{aligned} \tag{7}$$

Where $Z = -z + \frac{z^2}{2}(2+q_0)$.
Inserting (7) into (6):

$$\begin{aligned} \chi &= c\left(\frac{Z}{H_0}\right)\left(\frac{Z}{2} - 1\right) \\ \chi &= c\left(\frac{Z^2 - 2Z}{2H_0}\right) \\ Z^2 &= \frac{1}{4}(2+q_0)^2 z^4 + z^2 - (2+q_0)z^3 \\ &\text{neglecting } z^n, n > 2 \text{ terms:} \\ Z^2 &= z^2 \\ \therefore \chi &= c\left(\frac{z^2 - (2+q_0)z^2 + 2z}{2H_0}\right) \\ \chi(z) &= c\left(\frac{2z - (1+q_0)z^2}{2H_0}\right) \end{aligned} \tag{8}$$

Part d

Assuming the universe is flat $\kappa = 0$, thus $r(\chi) = \chi$. Then:

$$\begin{aligned} d_L(z) &= r(\chi)(1+z) = \chi(1+z) \\ d_L(z) &= c\left(\frac{2z - (1+q_0)z^2}{2H_0}\right)(1+z) \\ d_L(z) &= \frac{c}{H_0}\left(\frac{2z + 2z^2 - (1+q_0)(z^2+z^3)}{2}\right) \\ &\text{again neglecting } z^n, n > 2 \text{ terms:} \\ d_L(z) &= \frac{c}{H_0}\left(\frac{2z + 2z^2 - z^2 - q_0 z^2}{2}\right) \\ d_L(z) &= \frac{c}{H_0}\left(z + \frac{1}{2}z^2(1-q_0)\right) \end{aligned} \tag{9}$$

Question 2

Part a

$$d_L = \left(10^{\mu/5+1}\right) 10^{-6} \tag{10}$$

The speed of light is approximated as $c = 300000 kms^{-1}$.

$$\frac{d_L}{l} = \frac{100 d_L}{c} \tag{11}$$

Where the LHS $\frac{d_L}{l}$ is a dimensionless quantity.

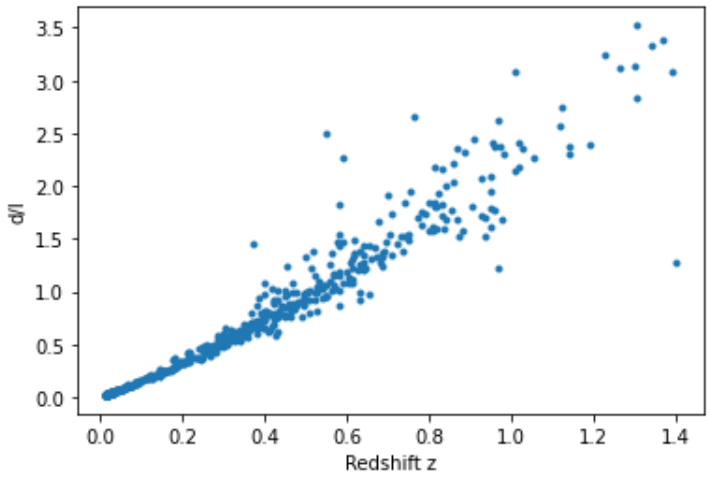


Figure 1: Graph of d_L/l against Redshift z.

Part b

Starting by restricting the values of the redshift and repeating the graph in Figure 1: Performing a linear

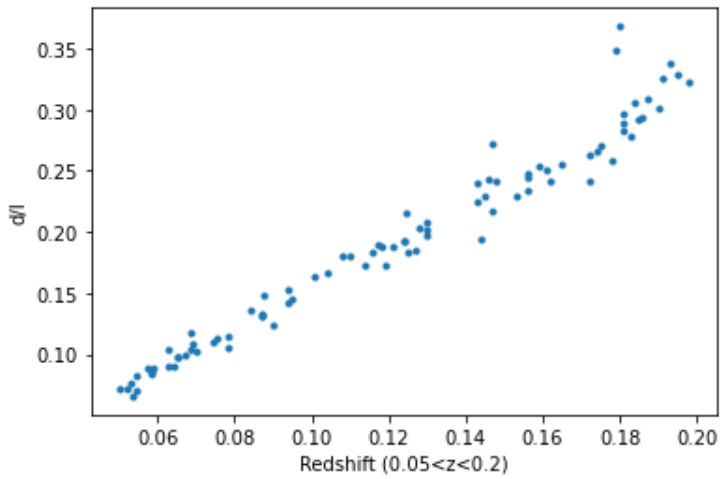


Figure 2: Graph of d_L/l against Redshift z for limited values.

regression test to estimate the intercept and slope of the graph returns the results seen in Figure 3 Using the

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Slope: 1.6977573747188222
Intercept: -0.014885097926152696
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Figure 3: Results of linear regression test on Figure 2.

results:
The linear regression returns the equation:

$$\frac{d_L}{l} = c_0 + c_1 z \tag{12}$$

Where c_0, c_1 are the intercept and slope values respectively. Then dividing both sides by z:

$$\frac{d_L}{zl} = \frac{c_0}{z} + c_1 \tag{13}$$

But by definition:

$$\begin{aligned} \frac{d_L}{z} &= \frac{c}{H_0} \\ \frac{d_L}{zl} &= \frac{1}{h} \end{aligned} \tag{14}$$

So equating (13) and (14):

$$\frac{1}{h} = \frac{c_0}{h} + c_1 \tag{15}$$

rearranging:

$$h(z) = \frac{z}{c_0 + c_1 z} \tag{16}$$

Using equation (16) and the values of c_0 and c_1 obtained from the regression, the values of $h(z)$ can be plotted for the limited range $z \in [0.05, 0.2]$. This can be seen in Figure 4 below. Current literature suggests the value

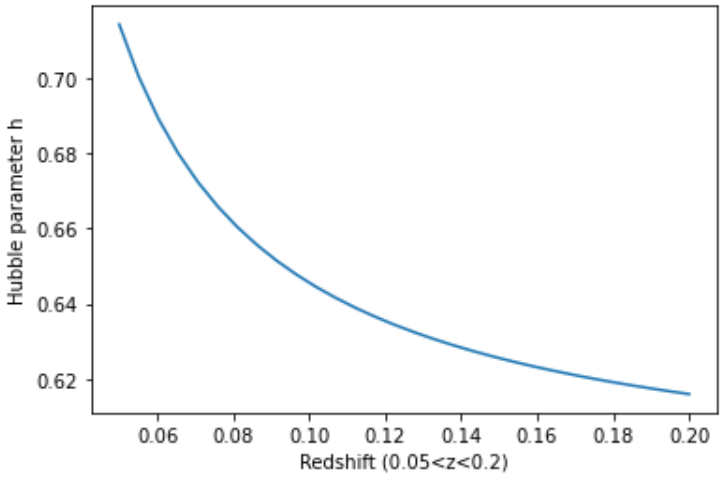


Figure 4: Graph of h against Redshift z for limited values.

of h is in the range $h \in (0.66, 0.75)$ with margins for error. The graph in Figure 4 suggests the value of h is in the range $h \in (0.62, 0.74)$, which is very similar to the results in current literature.

Part c

Now limiting the values of z to $z \in [0.1, 0.7]$ and replotting the graph in Figure 1:

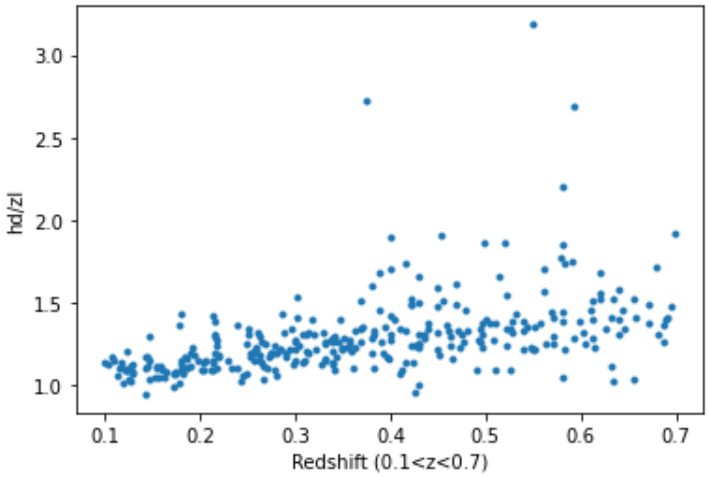


Figure 5: Graph of hd_L/zl against Redshift z for limited values.

Again performing a linear regression on the graph in Figure 5 to compute the intercept and slope for the equation:

$$\frac{hd_L}{zl} = c_0 + c_1 z \tag{17}$$

The results are shown in Figure 6

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intercept: 1.018169957500512
slope: 0.7339938430808871
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Figure 6: Results of linear regression test on Figure 5.

Using the equation derived in Question 1:

$$\frac{c}{H_0}\left(z + \frac{1}{2}z^2(1 - q_0)\right) \tag{18}$$

Then:

$$\begin{aligned} \frac{H_0 d_L}{c} &= z + \frac{1}{2}z^2(1 - q_0) \\ \text{But} \frac{H_0}{c} &= \frac{h}{l} \text{hence} \\ \frac{hd_L}{l} &= z + \frac{1}{2}z^2(1 - q_0) \\ \frac{hd_L}{lz} &= 1 + \frac{1}{2}z(1 - q_0) \end{aligned} \tag{19}$$

Equating equation (17) and (19):

$$\begin{aligned} c_0 + c_1 z &= 1 + \frac{1}{2}z(1 - q_0) \\ q_0(z) &= 1 - \frac{2(c_0 - 1 + c_1 z)}{z} \end{aligned} \tag{20}$$

Plotting the results for the limited range of z values:

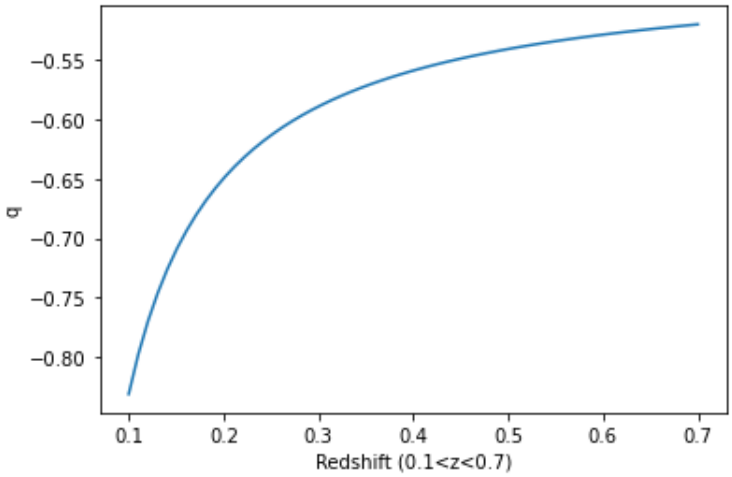


Figure 7: Graph of $q_0(z)$ against redshift z.

Current literature indicates the value of the deceleration parameter to be $q_0 \in (-0.9, -0.3)$, with a wide range of studies on the value resulting in many different estimates. The graph in Figure 7 suggests a value of $q_0 \in [-0.8, -0.55]$, which agrees with the literature. In this case, the range of values for q_0 is negative. This suggests that the universe will continue expanding at an accelerating pace.